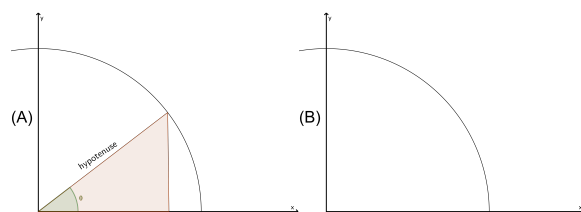


Problem 3

Problem 3

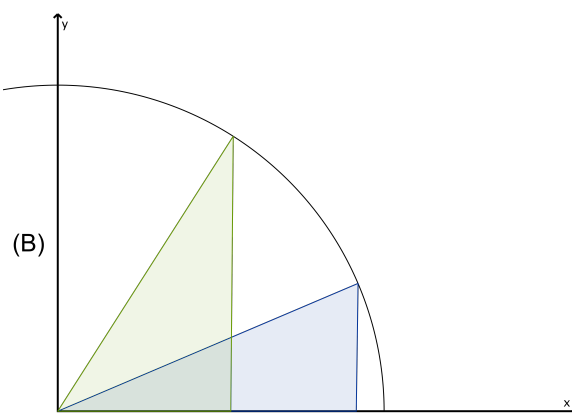
Problem 1 A part of a circle centered at the origin with radius $r = 7$ cm is given in the figure (A) below. A right triangle is formed in the first quadrant (see figure (A)). One of its sides lies on the x -axis. Its hypotenuse runs from the origin to a point on the circle. The hypotenuse makes an angle θ with the x -axis.



Make sure to label the picture.

- (a) Draw 2 more examples of such a triangle in the figure (B).

Explanation.



- (b) Express the area of such a triangle as a function of θ and state its domain.

Problem 3

Explanation. The area of a triangle is $(1/2) \cdot \text{base} \cdot \text{height}$. The base of the triangle is $7 \cos(\theta)$ and the height is $7 \sin(\theta)$.

Therefore:

$$\begin{aligned} A(\theta) &= \frac{1}{2} \cdot 7 \cos(\theta) \cdot 7 \sin(\theta) \\ &= \frac{49}{2} \cos(\theta) \sin(\theta) \end{aligned}$$

and

$$\text{Domain of } A = [0, \pi/2]$$

- (c) Find the value of θ which maximizes the area in part (b). Show your work and justify your answer. [2]

Explanation. Finding critical points:

$$\begin{aligned} A'(\theta) &= \frac{-49}{2} \sin(\theta) \sin(\theta) + \frac{49}{2} \cos(\theta) \cos(\theta) \\ \implies A'(\theta) &= 0 \\ \implies \cos^2(\theta) &= \sin^2(\theta) \\ \implies \cos(\theta) &= \sin(\theta) \\ \implies \theta &= \frac{\pi}{4} \end{aligned}$$

Locating global maximum:

$$A(0) = 0, A(\pi/4) = \frac{49}{4}, A(\pi/2) = 0 \implies \text{global maximum at } \theta = \pi/4$$